System design

FurFriends

System architecture, database (ERD) and flows

FurFriends is an installable web app where cat and dog owners in the Philippines keep their pets' health records, get care reminders, read first-aid guides, find nearby vet clinics, and ask the FurFriends AI, in English or Tagalog.

What owners can do

Keep profiles, medical history, vaccinations and documents for each pet; plan feeding and care with reminders; read first-aid and breed guides; find nearby vet clinics with hours and directions; ask the assistant.

How it is built

A React web app that works offline, a Supabase backend with private per-owner data, push reminders, and the team's own trained AI, running in a free Vercel function.

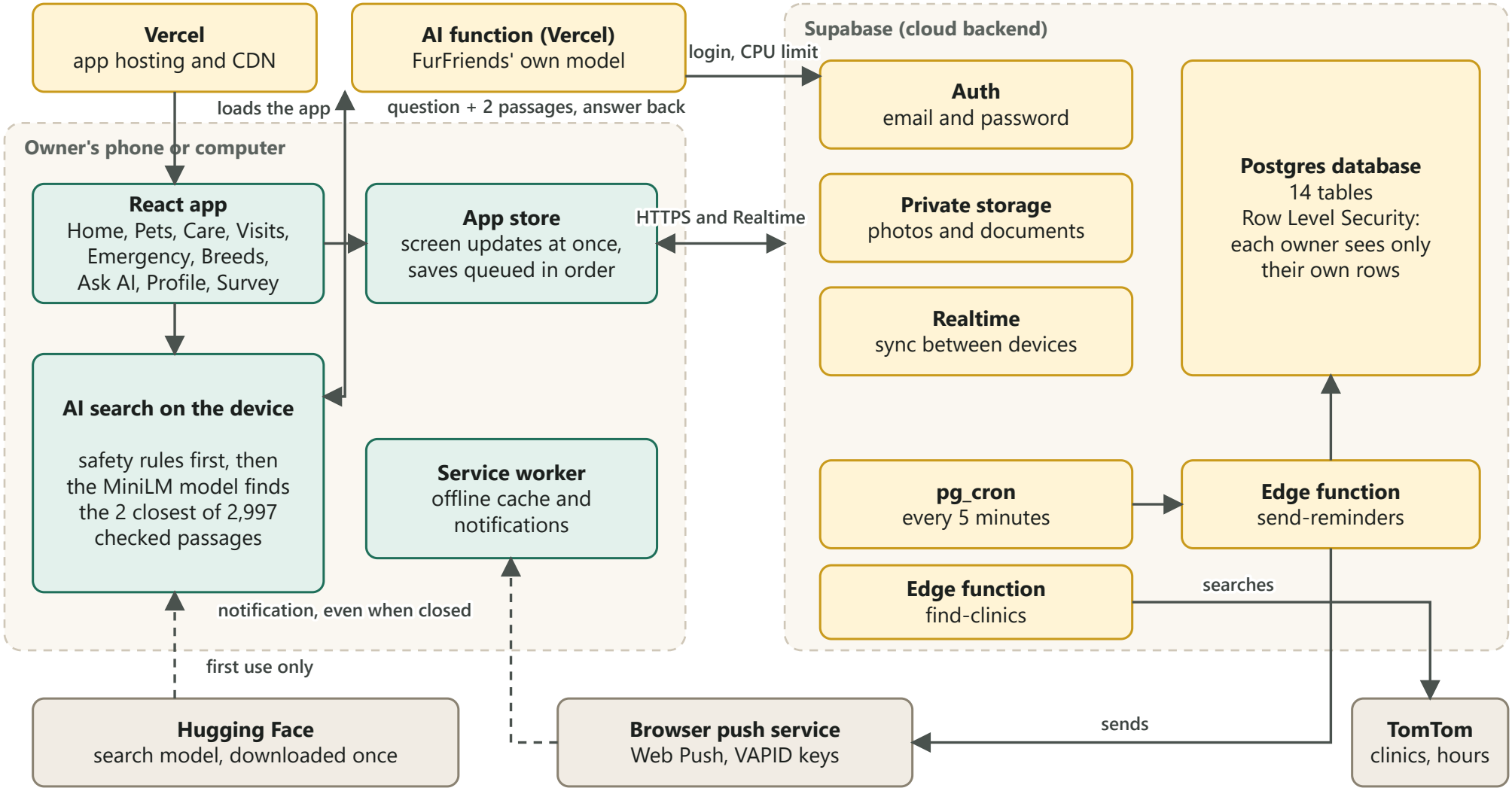
What this document shows

1. Architecture 2. Technology 3. ERD and data dictionary 4. Flows: sign-in and sync, the assistant, reminders, research data, deployment 5. Security and limits.

Live at furfriends-black.vercel.app. The database section matches supabase/migrations: 14 tables, 15 relationships (checked against the live project on 2026-10-24; ai_usage added 2026-10-28, clinic_search_cache 2026-10-29). Document date: 2026-10-29.

1. System architecture

Three places do the work: the owner's device (app, offline cache and the search), Vercel (serves the app and runs the AI that writes answers) and Supabase (accounts, data, files, sync, reminders and the AI's usage limit).



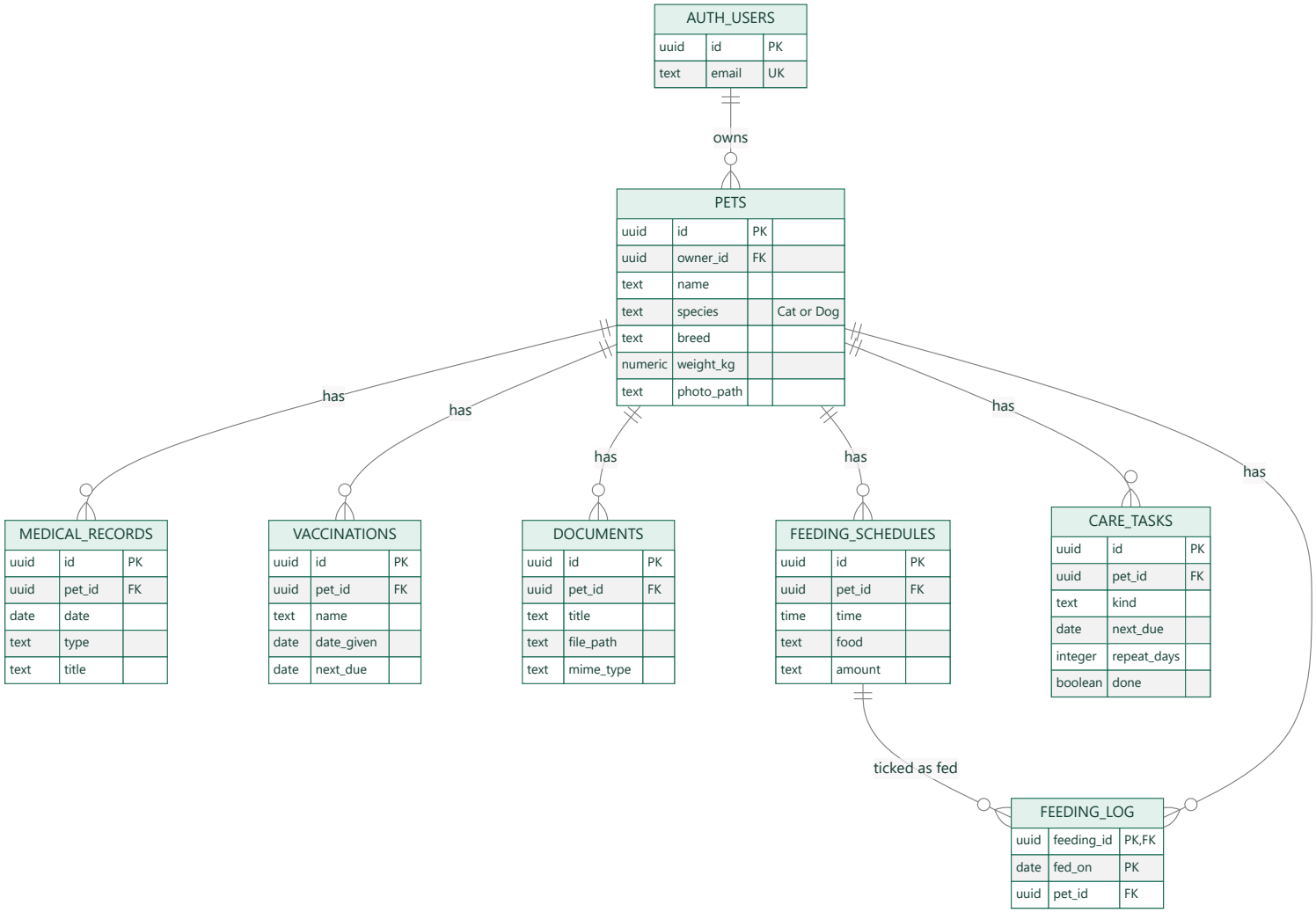
2. Technology and why

Each layer and the reason it was chosen.

Layer	Technology	Why
App	React 19, React Router, Vite	One codebase for phone and computer; fast to build and to test.
Installable and offline	PWA with a Workbox service worker	Works from the home screen; the app and answers keep working without a connection.
Accounts and data	Supabase Auth and Postgres with Row Level Security	Each owner can only read and change their own rows, enforced in the database itself.
Files	Supabase Storage (private buckets)	Photos and documents kept under the owner's folder, opened with signed links.
Sync between devices	Supabase Realtime	A change on one device reaches the others without a manual refresh.
Nearby vet clinics	TomTom Places Search API through a Supabase edge function (find-clinics); Leaflet map with OpenStreetMap tiles	Free with no card (5,000 searches a month), with opening hours for about half of Batangas clinics; the key stays secret; directions open Google Maps.
Reminders	pg_cron, an edge function, Web Push (VAPID)	Notifications arrive even when the app is closed, and are never sent twice.
AI search	Open-source MiniLM model in a Web Worker (transformers.js)	Finds the 2 checked passages closest to the question on the owner's device; it also works offline.
AI that writes the answer	FurFriends' own model (Qwen2.5-0.5B trained with LoRA, 4-bit ONNX) in a Vercel Python function, api/ask.py	Trained by the team on checked text; runs next to the app, so owners download nothing and there is no paid AI service.
Hosting	Vercel, deployed from GitHub	Every push to main publishes the new version.
Team tooling	Node scripts and Python (ai-server)	Builds the answer index, runs the accuracy test, content check, survey results and the secret scan.

3. Entity relationship diagram (1 of 2): owners, pets and their records

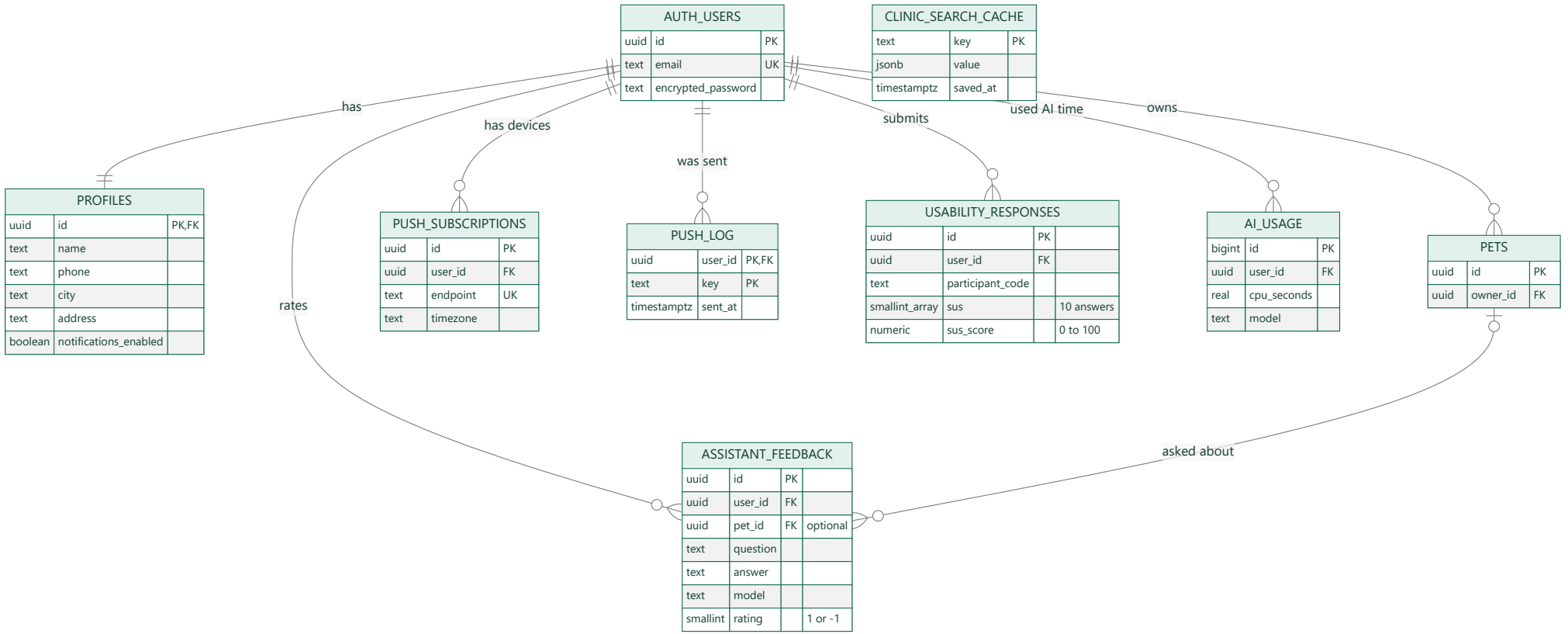
Every owner owns pets, and every health record, document, schedule and task belongs to a pet. Key columns are shown; the data dictionary lists every table.



PK = primary key, FK = foreign key, UK = unique. Deleting an account deletes its pets, and deleting a pet deletes everything under it.

3. Entity relationship diagram (2 of 2): accounts, reminders and research

The profile, reminder devices, survey answers and assistant ratings hang off the owner's login. A rating may also point to the pet it was about.



Ratings keep their question and answer if the pet is deleted (the pet link is cleared). One profile per login; the profile shares the login's id. clinic_search_cache stands alone: shared saved clinic searches that only the find-clinics function writes.

4. Data dictionary

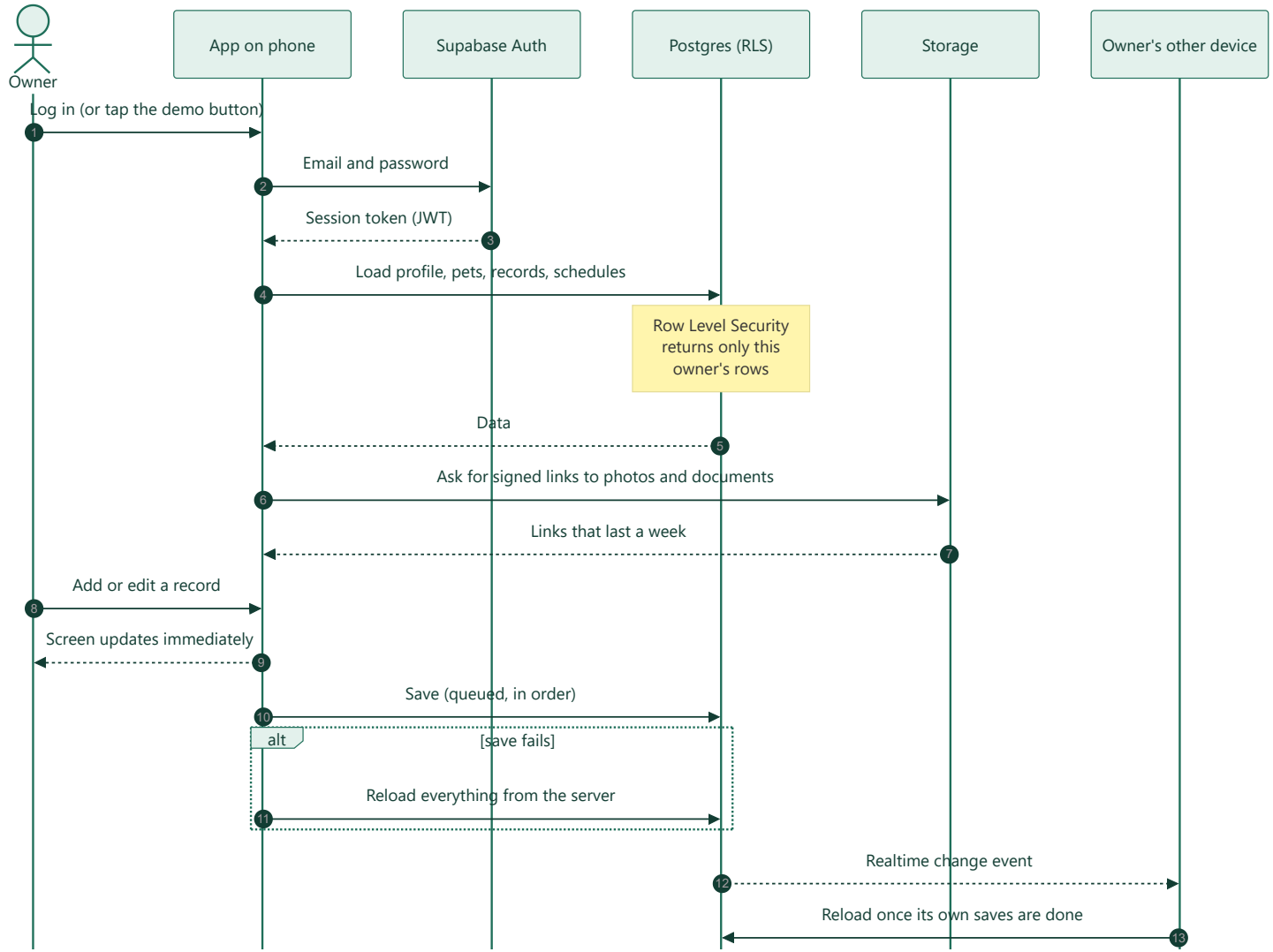
14 tables in the public schema, plus Supabase's own auth.users. Every table has Row Level Security on.

Table	One row is	Key	Rules
profiles	an owner's name, phone, city, address and reminder setting	id (the user's id)	One per user, created at sign-up.
pets	a cat or a dog	id	Species is Cat or Dog; age months 0 to 11; weight above 0.
medical_records	a checkup, illness, injury, surgery, medication or allergy	id	Belongs to a pet.
vaccinations	a vaccine given, with its next due date	id	Next due date is after the date given.
documents	an uploaded photo or PDF	id	The file is in private storage; the row keeps its path.
feeding_schedules	a meal time with food and amount	id	Days 0 to 6, empty means every day.
feeding_log	a meal ticked as fed on one day	feeding_id + fed_on	Only the last 14 days are loaded.
care_tasks	grooming, deworming, vaccination or a vet appointment	id	Repeat in days (0 = once); vet appointments are rows here, not a separate table.
assistant_feedback	a thumbs up or down on one assistant answer	id	Keeps the question, answer and model version; rating is 1 or -1.
push_subscriptions	a phone or browser that allowed notifications	id	Endpoint is unique; keeps the device's time zone.
push_log	a reminder already pushed	user_id + key	Stops the same reminder going out twice.
usability_responses	one tester's survey	id	Exactly 10 answers from 1 to 5; SUS score computed.
clinic_search_cache	one saved TomTom clinic search for a ~1 km area, or a typed address's location	key	Kept a week. Only the find-clinics function reads or writes it (no user policies); addresses are stored as a hash.
ai_usage	the CPU time one AI answer used	id	No question text. Users can add rows but not read them; the AI stops at 3 CPU-hours per 30 days.

Not in the database on purpose: the first-aid guides, breed guide and the assistant's answers are built into the app, so they work offline. There is no clinics table: nearby clinics come live from TomTom, and clinic_search_cache only keeps each search for a week.

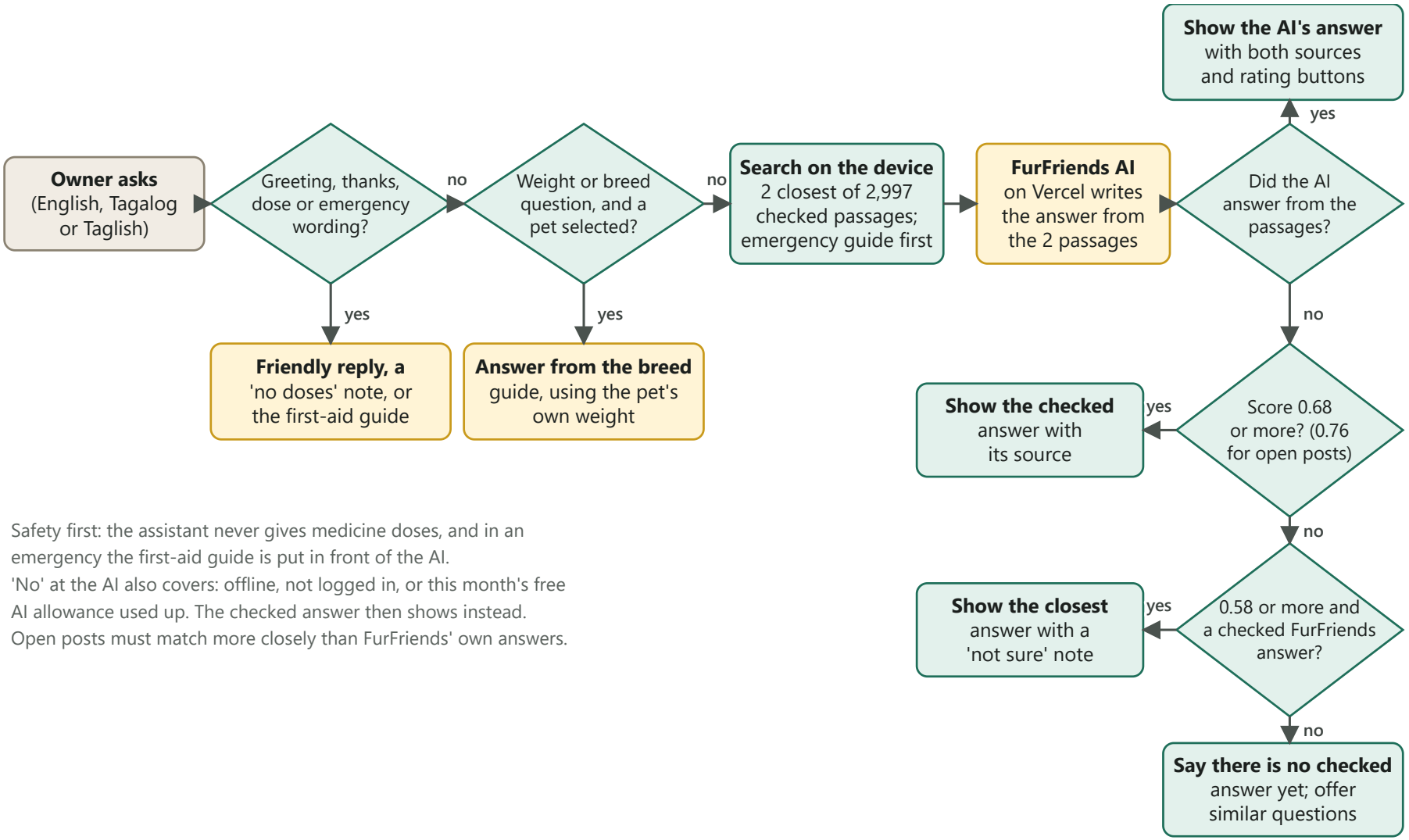
5. Flow: sign-in, loading and sync

How an owner's data reaches the screen, stays private, and appears on a second device.



6. Flow: asking the assistant

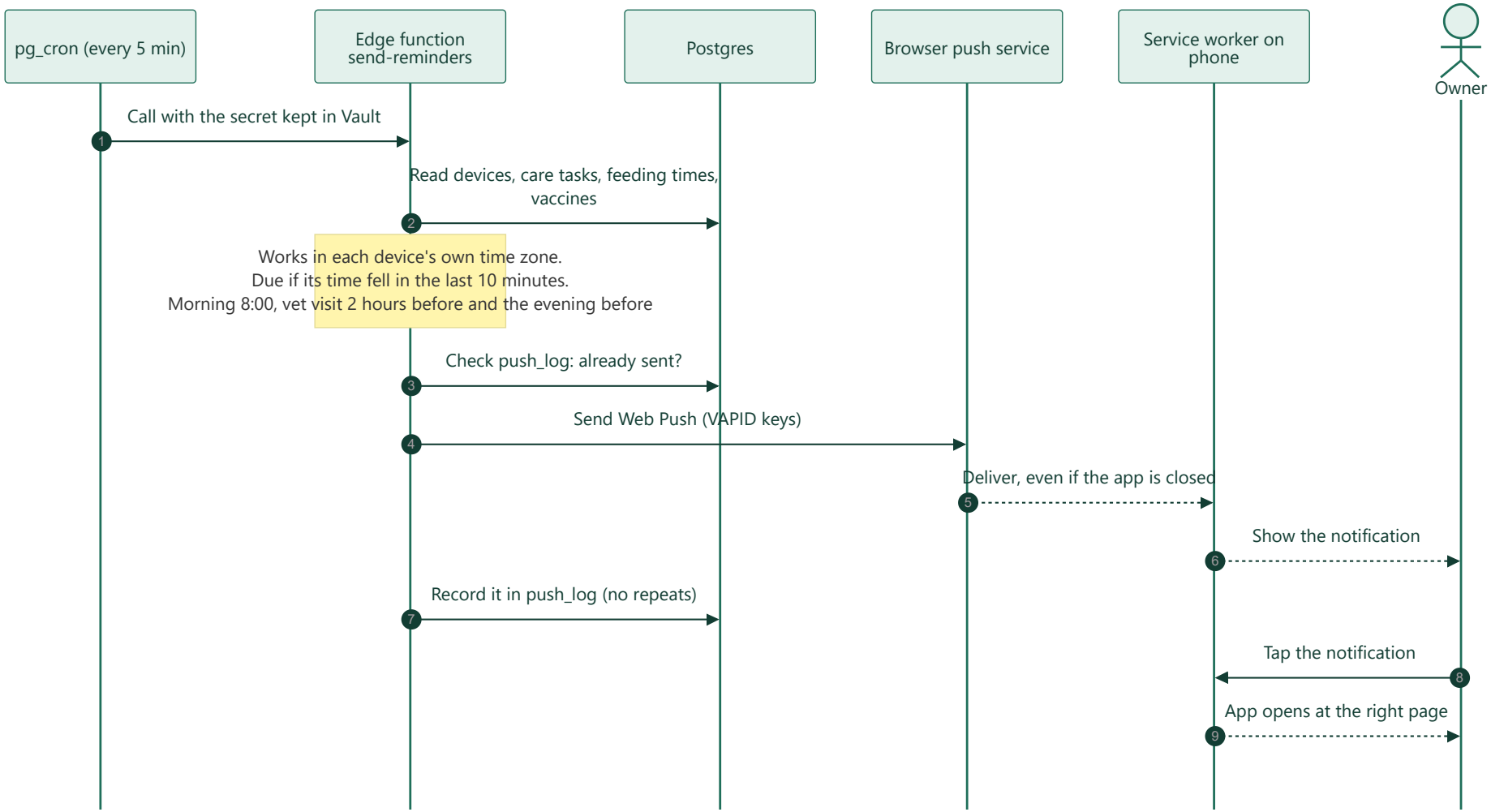
Rules run first. Then the device finds the 2 closest checked passages, and FurFriends' own model on Vercel writes the answer from them. If it can't, the closest checked answer shows, only when it is close enough.



Safety first: the assistant never gives medicine doses, and in an emergency the first-aid guide is put in front of the AI.
'No' at the AI also covers: offline, not logged in, or this month's free AI allowance used up. The checked answer then shows instead.
Open posts must match more closely than FurFriends' own answers.

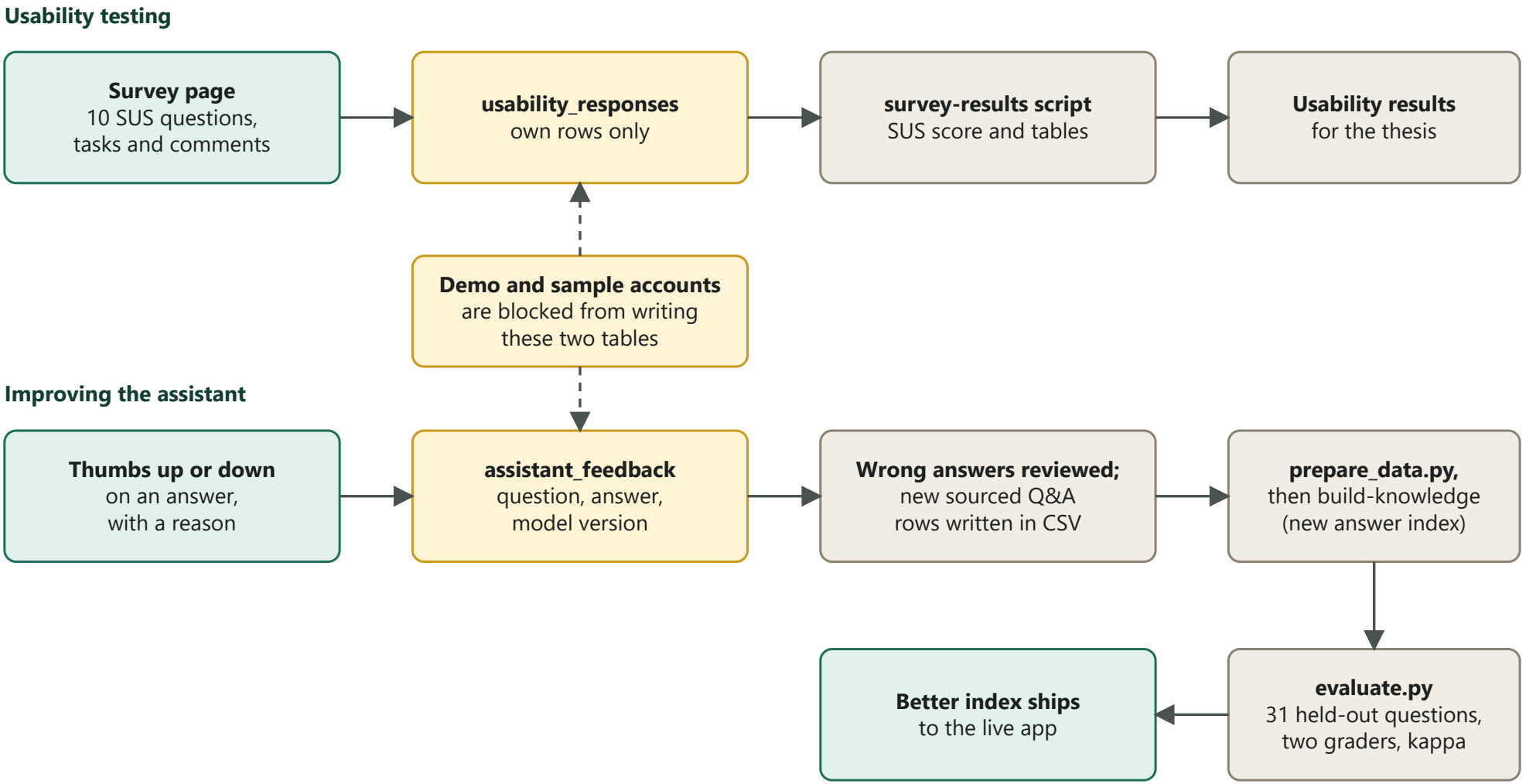
7. Flow: reminder notifications

A schedule in the database wakes an edge function every 5 minutes. Reminders arrive even when the app is closed.



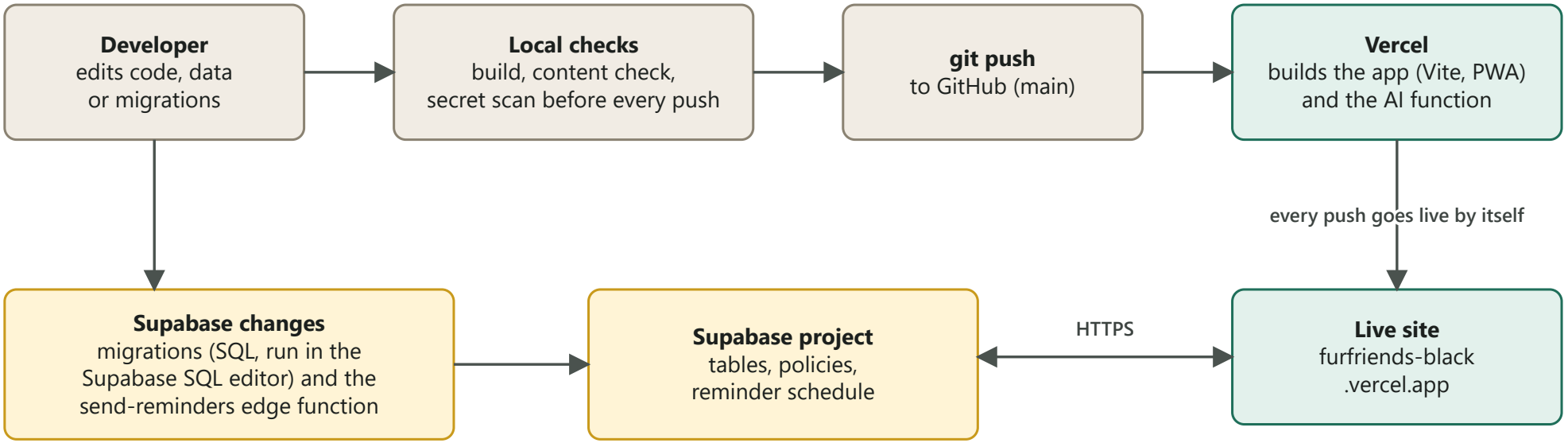
8. Flow: research data and improving the assistant

Survey answers and ratings are collected from real testers only, then used to measure usability and to fix wrong answers.



9. Flow: deployment

Code and data go through local checks, then GitHub, then Vercel. Database changes are SQL migrations applied to Supabase.



Database changes are written as numbered SQL files in supabase/migrations and applied in order.
The reminder schedule and push keys live in Supabase (Vault and edge function secrets), never in the app.

10. Security, privacy and limits

Area	How it is handled
Each owner sees only their own data	Row Level Security on all 14 tables; a pet's rows are visible only when its owner is the logged-in user. Visitors who are not logged in see nothing.
Files are private	Photos and documents sit under a folder named with the owner's user id, locked by a storage policy, and are opened with links that expire after a week.
Secrets stay on the server	The app holds only the public Supabase key. The service-role key and push private key live in the edge function's secrets. A secret scan runs before every push and passed on the whole history.
Locations aren't kept	The phone's location is used for one search and never saved. Searches are saved per ~1 km area, and a typed address only as a hash. The TomTom key is an edge-function secret, and only logged-in owners can search.
Questions go only to FurFriends' own AI	Only logged-in owners can use it. The question and 2 checked passages go to the team's function on Vercel, not to a paid AI service, and are not stored. A question is shared with the team only when the owner rates the answer.
The free plan can't be overrun	Each answer's CPU time is logged in Supabase. Above 3 of the free 4 CPU-hours per 30 days, the AI stops and the checked answer shows instead, so the app is never paused.
Shared demo accounts are protected	The demo and sample logins have public passwords, so the database refuses changes to their password or email, and refuses their survey answers and ratings.
The assistant is cautious	It writes only from 2 checked passages, never gives medicine doses, puts the first-aid guide first in an emergency, shows its sources, and says when the passages don't answer the question.

Limits to state honestly

- The AI is a small model (0.5 billion parameters). It writes from the 2 closest checked passages but can still pick the wrong ones or word an answer wrongly, and the screen says so. It is not a vet.
- There is no vet adviser. Answers are written from public sources (VCA, Merck, Cornell, ASPCA, WSAVA, Philippine laws) and checked against them by the team, which is a limitation to state.
- Tagalog and Taglish work through a word list that turns common words into English before the search. Unusual wording can still be missed.
- The search model (about 30 MB) downloads once, on the first question. Writing answers needs internet; offline, the closest checked answer shows instead.
- The AI shares Vercel's free plan: about 25 to 50 CPU-seconds per answer, so roughly 200 to 400 answers per 30 days before it falls back to checked answers.
- Clinic listings and hours come from TomTom and can be wrong or out of date; about half of Batangas clinics have no hours listed, and small towns have few listings. The app says to call before going.
- Survey and grading results shown anywhere as SIMULATED are practice data only, not research results.